

Vision-aided Quaternion Filter for Robust 3D Wrist Orientation Estimation

Extending existing filters with accelerometer, magnetometer and marker-based vision corrections

Research Question

- How can a quaternion filter use asynchronous and intermittent vision updates while retaining low-latency inertial tracking and robustness to magnetic disturbance?

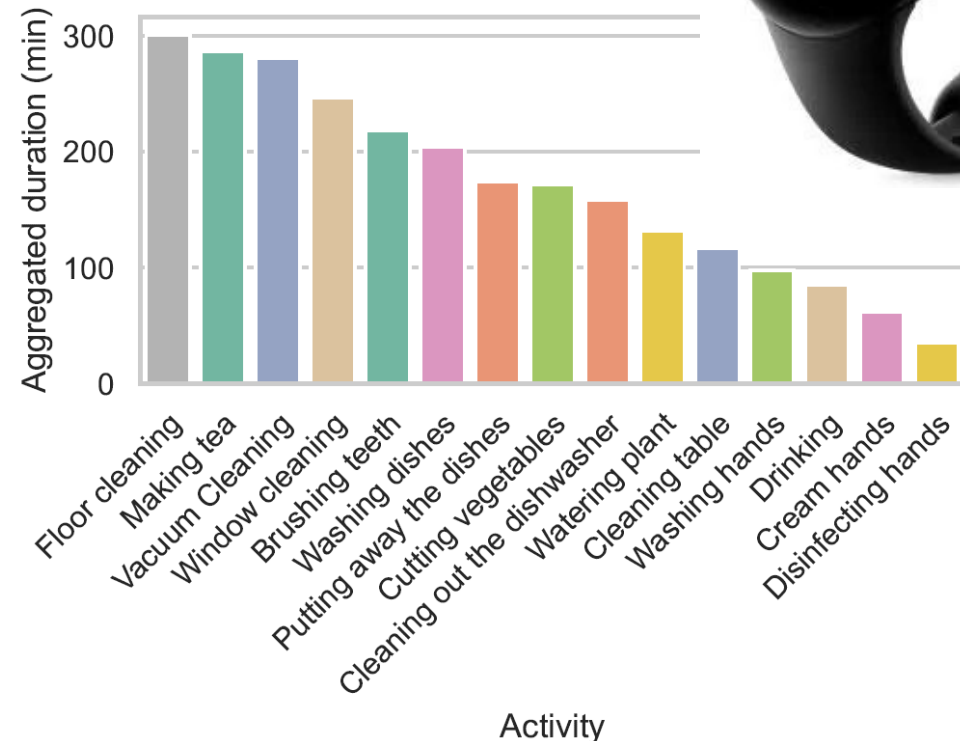
Thesis Scope

- Review and benchmark existing filters and vision-aided filters.
- Design confidence-weighted 3D vision correction and explicit timing and frame alignment.
- Handle magnetic disturbance, marker ambiguity, occlusion and delayed measurement
- Implement the real-time filter and validate it against Vicon ground truth using a reproducible benchmark.



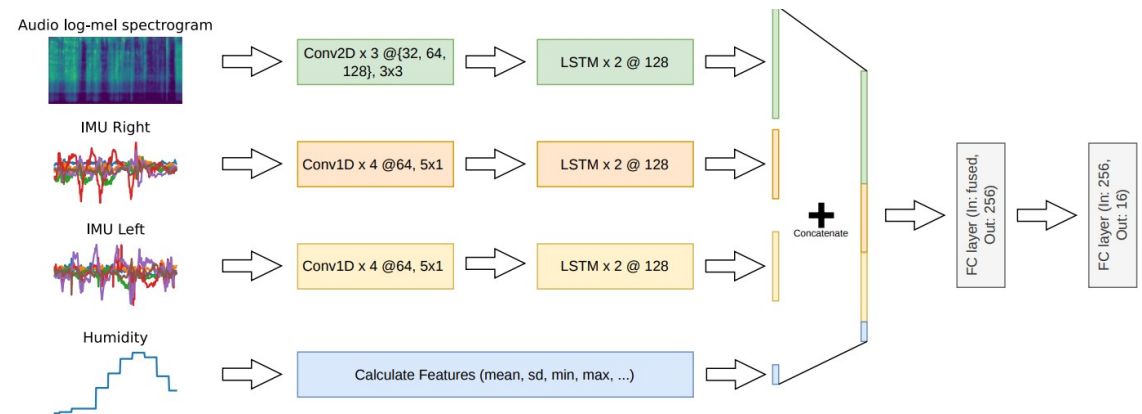
Deployment and evaluation of a wearable multi-modal HAR system on the HARMES dataset (WearOS)

- Start Date: ~1st Nov. 26
- Existing HAR dataset
- Model for WearOS and implementation must be created
- Run user study for online activity recognition
- Smartwatch provided
- Android / Pytorch / Deep Learning background recommended



Self-supervised pre-training of multi-modal HAR models on the HARMES dataset

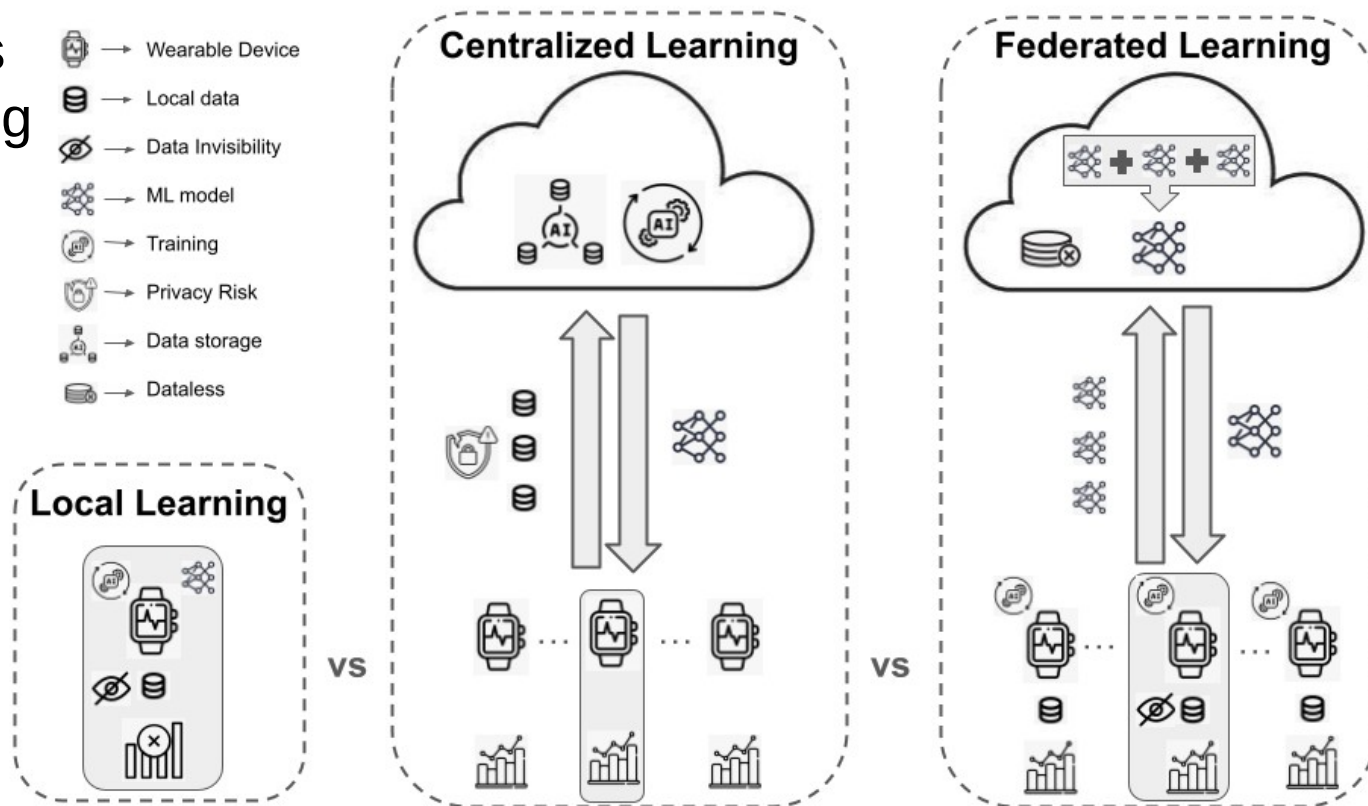
- Start Date: ~1st Nov. 26
- Existing 80h HAR dataset (~60h labeled)
- Use non-scripted part for pre-training
- Select pre-training mechanism and compare to baselines
- Data provided
- Pytorch / Deep Learning background recommended



Pre-train encoder → Fine-tune on actual task →
Evaluate vs non pre-trained models.
Optional: Check transfer performance on other
datasets (e.g. WEAR)

Enabling Trustworthy Collaborative Learning Systems for Human Activity Recognition

- Development of ML systems for the HAR domain by taking GDPR measures.
- Enhancing Federated Learning considering the characteristic of the target application.
- Evaluating metrics of designed system with existing learning systems using wearable device data.



Towards A Medical-Grade Long-Term Activity Sensing Widget

- Develop an embedded Widget that fits in just a few kilobytes and operates in the Bangle open-source smartwatch ecosystem
- Design lossless compression & capture algorithms to acquire and save accurately-timed sensor data from motion and heart rate sensors
- Test your setup on a custom testbench

More information in
H-A8110 or: kvl@eti.uni-siegen.de

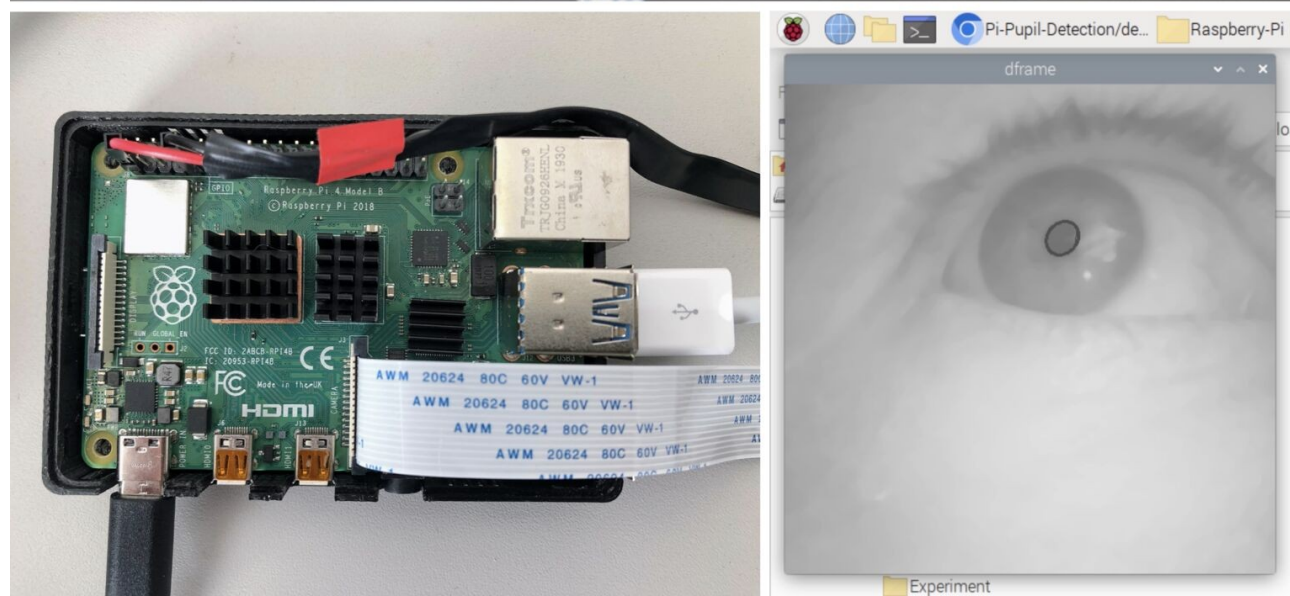
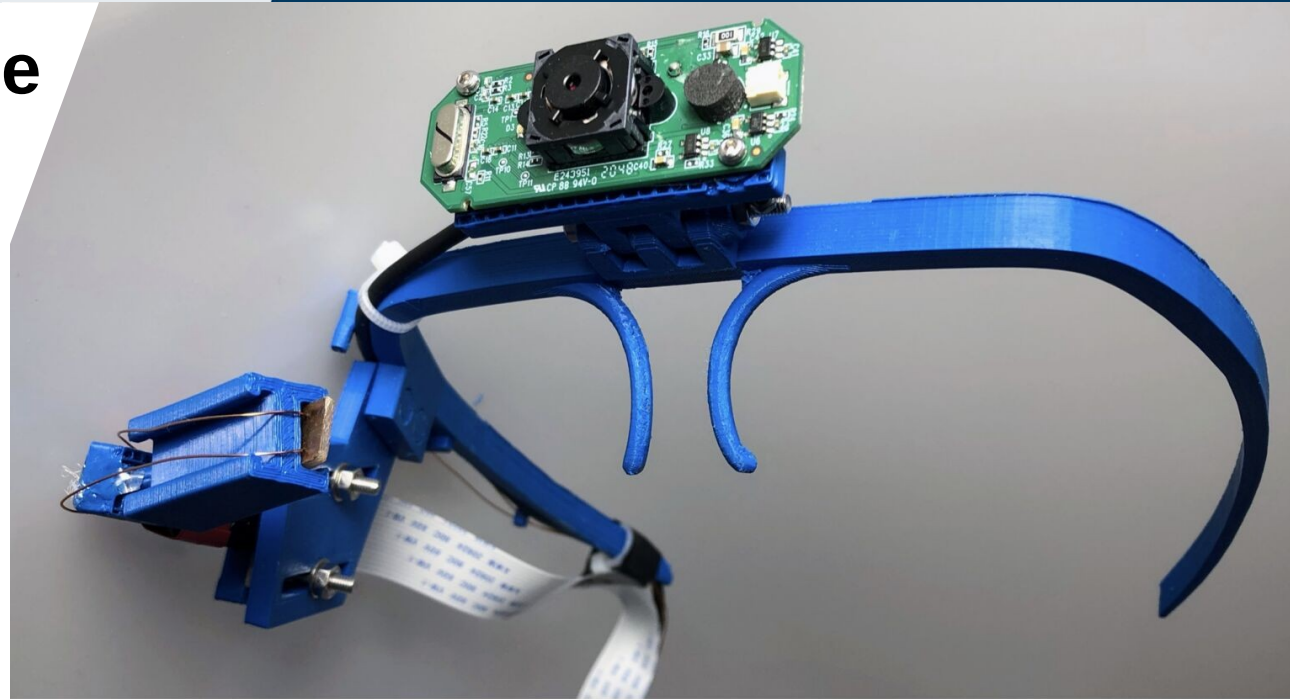


Thesis Topic

An Open-Source & Realtime Embedded Eye-Tracker

- Development of a Python and OpenCV-based wearable system on a Raspberry Pi
- Based on our existing pupil detection system
- Evaluation with two existing state- of-the-art eye tracking systems

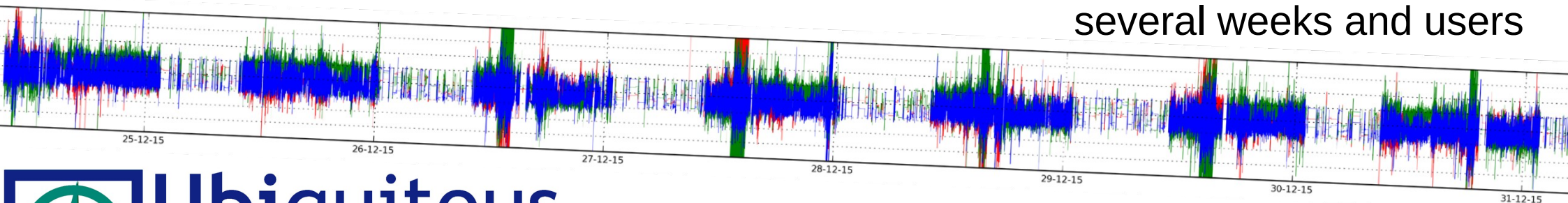
More information in **H-A8110**
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A Wearable Stairs vs. Elevator Detector

- Using the internal barometric pressure and inertial sensor data from the Bangle.js.v2 watch, to detect when the wearer uses the stairs or lift
- Development of an efficient embedded algorithm in the Espruino IDE, an embedded open-source ecosystem to enable long-term operation
- Goal should be to deploy and validate the algorithms in a longitudinal deployment study of several weeks and users



Updating the process to record entry's of the WEAR Dataset

- Implementing Bluetooth synchronisation
- Implementing approximation saving
- Creating a more user-friendly way to convert data
- Compare recording between old and new

More information in H-A 8115 or
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Improvement of workflow and usability of ChronoVault

- Develop a workflow to label Data easier in the wild
- Improve the way to copy data from ChronoVault
- Make it easier to configure the recording (Resolution, Rotation,...)
- Goal: People outside of CS should be able to use ChronoVault

More information in **H-A 8115** or
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How attentive are passengers on Public transport

- Activities on public transport are encouraged, though how efficient is it?
- Combine a Bangle with Esense
- Measure interruptions of tasks and compare it to other places (e.g. Office)

More information in **H-A 8115** or
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